

LLM-Powered Multi-Agent Negotiation in Buyer–Seller Markets: Simulator Design, Experimental Evaluation, and Methodological Analysis

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Abstract

TEST TEST We present a configurable multi-agent negotiation simulator for evaluating LLM bargaining behaviour in alternating-offer bilateral negotiation under private economic constraints, and report three contributions: the simulator itself, a structured experimental evaluation, and a methodological analysis of how simulator design choices affect observed outcomes.

Using a single 3B-parameter language model (`llama3.2:3b`), we conduct seven experiments totalling 4,120 negotiation sessions across three levels of analysis: micro-level bargaining dynamics (concession curves, anchoring bias, deadline pressure), macro-level market outcomes (price discovery, stochastic shock response, structural supply-demand shifts), and institutional mechanism effects (matching algorithm comparison).

Several platform-level outcomes are directionally consistent with classical bargaining and market theory: deal rates scale with the width of the zone of possible agreement, the buyer-first protocol produces systematic surplus asymmetry, structural supply-demand shifts produce directionally correct price responses with near-complete cost pass-through on the supply side (+17.9% from a +\$20 cost shift across three seeds), and surplus-maximising matching improves deal rates by 16 percentage points and allocative efficiency by 5.8 percentage points relative to random matching. In single-seed exploratory runs, market prices show no systematic trend over 30 ticks and exogenous shocks reduce prices and liquidity in theoretically predicted directions.

However, a configurable feasibility gate—necessary to prevent small-model output failures—drives approximately 40% of all acceptances (and effectively all acceptances in the micro-level experiments), meaning that many positive outcomes reflect gate enforcement and protocol structure rather than agent-level decision-making. Agents do not exhibit two well-documented human bargaining heuristics—anchoring bias and deadline-driven concession acceleration—but these null results are heavily confounded by the gate (which settles deals before the relevant dynamics

can manifest) and by prompt design choices, making it unclear whether the absences reflect LLM cognition or experimental design constraints.

A central methodological finding is that simulator design choices—the feasibility gate, prompt architecture, and protocol structure—function as first-order experimental variables that materially alter the apparent economic behaviour of LLM agents. The platform makes these choices explicit, quantifies their involvement, and logs every gate-fired settlement separately, enabling assessment of how much observed behaviour is agent-driven versus mechanically enforced. The empirical results characterise one model under one prompt architecture and should not be generalised without gate ablation, prompt ablation, and multi-model comparison.

1 Introduction

1.1 Background

Large language models are increasingly deployed as autonomous agents that interact with environments, tools, and other agents. Recent work has shown that LLM-powered agents can exhibit emergent social behaviours in open-ended simulation [11], coordinate within multi-agent frameworks [19], and interleave reasoning with action [20]. A natural extension of this research agenda is to ask whether LLM agents can act as economically meaningful negotiators — that is, whether they produce bargaining behaviour that is interpretable through the lens of established economic theory.

Bilateral bargaining is among the best-characterised settings in economic theory. Rubinstein’s alternating-offer model [14] predicts that patient agents extract larger surplus shares through time-preference discounting. The anchoring literature [16] establishes anchoring as a general cognitive heuristic; in negotiation contexts, Galinsky and Mussweiler [5] show that first offers systematically bias final agreements. Deadline pressure effects have been documented in human experiments [13], and decentralised bilateral markets are expected to produce equilibrium prices with persistent price dispersion [15]. These theoretical predictions provide clear, testable benchmarks for evaluating agent behaviour.

Despite growing interest in LLM negotiation, existing studies typically focus on multi-issue dialogue games [8, 1], personality-driven interaction [9], or assistive human–AI bargaining [3]. These designs evaluate negotiation capability but do not systematically test LLM agents against the specific quantitative predictions of classical bargaining theory. Furthermore, most studies operate at a single level of analysis without connecting micro-level bargaining outcomes to macro-level market dynamics within a single controlled platform. The practical significance of this gap is growing: automated negotiation is entering consumer and enterprise markets [4], and understanding where LLM behaviour aligns with or departs from economic theory is an important input for anticipating the consequences of deployment.

1.2 Problem Statement

The central question motivating this thesis is:

Can LLM-based autonomous agents serve as economically meaningful negotiators in a language-mediated buyer–seller market simulator, and under what simulator and protocol designs do meaningful micro- and macro-level patterns emerge?

This question cannot be answered by evaluating LLM outputs in isolation. Simulator design choices — the feasibility gate that enforces settlement, the prompt architecture that structures agent reasoning, and the protocol design that governs turn-taking and termination — all materially alter the observable behaviour of agents. A feasibility gate that deterministically accepts profitable offers, for example, may suppress the very dynamics (late concessions, deadline pressure) that an experiment is intended to measure. These design choices are therefore first-order experimental variables, not neutral background conditions.

A second challenge is attribution. Any observed outcome in an LLM simulation reflects the combined system: model, prompt, gate, and protocol. Disentangling the contribution of LLM decision-making from the contribution of simulator infrastructure requires careful logging, explicit ablation studies, and conservative interpretation. This thesis builds a platform designed to support such analysis, though it does not fully resolve the attribution problem within the scope of a single project.

1.3 Research Objectives

This thesis pursues four objectives:

1. Design and implement a configurable multi-agent negotiation simulator that enforces hard economic constraints, supports multiple agent types, and provides structured event logging sufficient to distinguish gate-driven from LLM-driven settlements.
2. Conduct a systematic evaluation across seven experiments covering concession dynamics, anchoring bias, deadline pressure, market price discovery, shock response, mechanism design, and supply–demand responsiveness.
3. Characterise, with appropriate conservatism, where platform outcomes are directionally consistent with classical theory and where they depart from it, distinguishing genuine LLM behaviour from gate-induced artefacts wherever the evidence permits.
4. Analyse how simulator design choices — including the feasibility gate, prompt structure, and matching mechanism — function as experimental variables that shape the apparent behaviour of agents.

1.4 Contributions

This thesis makes three contributions. All results characterise the combined system (LLM + rethink + gate + prompt) under specific conditions; they should not be generalised across model families or scales without further ablation and replication.

Simulator. We present a configurable multi-agent negotiation platform for buyer–seller bilateral bargaining. The platform enforces private constraints (buyer value, seller cost, budgets), implements an alternating-offers protocol with configurable round horizons, and applies a deterministic feasibility gate that accepts clearly profitable offers without calling the LLM. LLM outputs pass through an internal rethink loop that corrects parse and self-constraint errors before they reach the session’s gate and ActionJudge. Every gate intervention is logged so that gate-driven and LLM-driven settlements can be distinguished in analysis. A market-level execution layer connects individual negotiation sessions to aggregate outcomes, enabling simultaneous analysis of micro-level bargaining dynamics and macro-level price and liquidity statistics. Three agent types are supported: a deterministic rule-based baseline, an LLM reactive agent, and an LLM deliberative agent whose structured reasoning draws on chain-of-thought prompting [17].

Experimental evaluation. We conduct seven experiments totalling 4,120 negotiation sessions using a single model (llama3.2:3b via Ollama at temperature 0.2). Experiments A–C (concession curves, anchoring, deadline pressure) are replicated across three seeds; Experiments D and E (market price discovery and stochastic shock response) are single-seed exploratory runs whose results carry higher uncertainty; Experiments H and I (mechanism design, supply–demand shifts) are replicated across three seeds. Several results are directionally consistent with economic predictions: deal rates scale with zone-of-possible-agreement (ZOPA) width (2.7% at ZOPA = \$20 versus 100% at ZOPA ≥ \$40), a pattern consistent with gate-enforced feasibility constraints as well as agent behaviour. Structural supply shocks are associated with near-complete cost pass-through (+17.9% price from a +\$20 cost shift across three seeds), and surplus-maximising matching improves deal rates by 16 percentage points and allocative efficiency by 5.8 percentage points relative to random matching. Two prominent null results — no anchoring effect ($r = -0.148$ between first offer and final price) and no deadline-driven concession acceleration — are substantially confounded by the feasibility gate and prompt design, and should not be interpreted as evidence about LLM cognition in general.

Methodological analysis. We provide an analysis of how simulator design choices function as first-order experimental variables. The feasibility gate drives approximately 40% of all acceptances globally and effectively 100% of acceptances in Experiments A–C, making it difficult to attribute positive outcomes in those experiments to LLM decision-making without gate ablation. Prompt architecture also has measurable effects: the deliberative agent produces seller counter-offers approximately \$7 lower than the reactive agent on the first counter, a difference consistent with a prompting effect, though al-

ternative explanations cannot be excluded without controlled ablation. These findings suggest that the design of the simulation infrastructure is not separable from the findings it produces — a methodological concern that may extend to other efforts to use LLM agents as proxies for human economic behaviour.

Multi-model comparison and gate ablation at the experiment level are identified as priorities for future work.

The remainder of this thesis is organised as follows. Section 2 reviews related work in LLM-based simulation, LLM negotiation, and classical bargaining theory. Section 3 describes the problem formulation and simulator architecture. Section 5 details the methodology and experimental design for all seven experiments. Section 6 presents results and analysis. Section 7 discusses implications, limitations, and the role of simulator design choices as experimental variables. Section 8 concludes and identifies directions for future work.

2 Literature Review

This work sits at the intersection of agent-based simulation, LLM-based agent design, and computational bargaining research. We survey these threads and identify the gap that motivates the platform.

2.1 Agent-Based Modeling and LLM-Based Simulation

Agent-based modelling (ABM) has long been used to study emergent economic phenomena from micro-level interactions, including market price formation, information diffusion, and collective behaviour [6]. Traditional ABM agents follow scripted rules or simple utility functions, which limits behavioural realism. The advent of large language models has opened a new direction: using LLMs as behavioural engines, so that agents respond to context in natural language and exhibit richer, less predictable strategies.

Several systems have demonstrated this potential at scale. S3 [7] simulates social-network dynamics with LLM agents, studying opinion formation and information spread. AgentSociety [10] scales LLM-driven agents to large populations, capturing emergent social and economic phenomena. Williams and Hosseinmardi [18] apply generative agents to epidemic modelling, showing that LLM agents can reproduce stylised public-health dynamics in structured environments. Gao et al. [6] survey the broader LLM-ABM landscape and identify economic simulation as a high-potential but under-explored domain. Our platform contributes to this domain by situating LLM agents within a bilateral bargaining protocol under hard economic constraints, enabling direct comparison against classical predictions.

2.2 Generative Agents and Agent Architecture

Park et al. [11] introduced *Generative Agents*, demonstrating that LLM-powered characters in a sandbox environment exhibit emergent social behaviours—forming

relationships, spreading information, and coordinating activities—without explicit scripting. This established LLMs as plausible behavioural engines for simulation, though the focus was on open-ended social dynamics rather than constrained economic decision-making.

At the architectural level, the ReAct framework [20] introduced a reasoning-and-acting paradigm in which models interleave chain-of-thought reasoning with environment actions. Our deliberative agent design adapts this pattern for structured negotiation: the agent reasons over its private constraints before producing a priced offer. Multi-agent orchestration frameworks such as AutoGen [19] provide general infrastructure for LLM agent interaction but target open-ended tool use rather than controlled economic experiments. Our platform differs from these systems by embedding agents within a fixed alternating-offer protocol with hard feasibility constraints, producing behaviour that can be evaluated against quantitative theoretical baselines.

2.3 LLM Negotiation Literature

Negotiation is among the earliest testbeds for learned agents. Lewis et al. [8] trained neural negotiation agents via reinforcement learning and self-play on a multi-issue bargaining task, showing that learned models reach agreements but can degenerate into non-human-like strategies. That work predates LLM-based approaches and uses task-specific training; our agents are prompted, not trained, and negotiate a single price under private constraints rather than trading across multiple issues.

Classical bargaining theory provides the normative benchmarks against which LLM behaviour is evaluated throughout this paper. Rubinstein’s [14] alternating-offer model predicts that a more patient agent extracts a larger share of surplus; we test whether LLM agents reproduce this effect. The anchoring literature, originating with Tversky and Kahneman [16] and refined in negotiation contexts by Galinsky and Mussweiler [5], predicts that first offers bias final agreements. Roth, Murnighan, and Schoumaker [13] established that human bargainers make disproportionately large concessions near deadlines; we test this prediction by varying time horizons. At the market level, Rubinstein and Wolinsky [15] showed that decentralised bilateral bargaining produces persistent price dispersion, predicting that the Law of One Price fails under bilateral matching. Raiffa’s [12] concept of the zone of possible agreement (ZOPA) is central to our feasibility gate and to the observation that ZOPA width strongly predicts deal success in our experiments. On the computational side, Zheng et al. [21] demonstrate that multi-agent reinforcement learning can discover effective economic policies in simulated economies, motivating the use of AI agents for mechanism-design experiments.

With the advent of large language models, several works have studied LLM behaviour in richer negotiation settings. Abdelnabi et al. [1] evaluate LLMs in interactive multi-agent negotiation games, finding that deliberation-style prompting improves agreement quality. Their multi-issue, multi-party design complements our single-issue bilateral setting, which isolates price dynamics from

issue-trading effects. Mao et al. [9] show that personality prompts measurably shift negotiation outcomes, motivating our communication-strategy ablation. Chawla et al. [3] develop assistive LLM agents for socially-aware dialogue, emphasising cooperative and empathetic communication; our agents negotiate autonomously with structured JSON actions, prioritising experimental control over dialogue realism. Amato et al. [2] simulate LLM-based negotiators within a game-theoretical framework, and Davidson et al. [4] model agent-to-agent negotiations in consumer markets, both examining richer environments than ours. The chain-of-thought prompting literature [17] underpins the deliberative reasoning pattern used in our platform.

2.4 Research Gap

The literature reviewed above leaves three gaps that motivate this work. First, existing LLM simulation platforms operate primarily at either the micro level—studying individual agent behaviour and reasoning—or the macro level—studying aggregate social or economic outcomes—but no single platform systematically connects both within a controlled experimental design. Second, while classical bargaining theory provides precise quantitative predictions, no prior work has tested these predictions against LLM agents in a systematic, reproducible manner; most studies report qualitative or task-specific findings rather than comparing against theoretical benchmarks such as Rubinstein equilibrium splits or deadline-effect magnitudes. Third, the influence of simulator design choices—protocol structure, feasibility enforcement, prompt formulation, matching mechanism—on observed behaviour has received little attention, leaving open the question of how much reported LLM negotiation behaviour reflects genuine agent cognition versus experimental confounds.

Our platform is designed to address these gaps. It connects micro-level bargaining dynamics and macro-level market outcomes within one system, enabling the causal chain from individual offers to aggregate price distributions to be traced. It provides quantitative comparisons against classical predictions across seven experiments, reporting both confirmations and null results. It also makes simulator design choices explicit and ablation-testable, so that the effect of feasibility gating, communication strategy, and matching mechanism can be isolated. The contribution is methodological and empirical rather than algorithmic: we offer a reproducible testbed with hard economic constraints and deterministic fallbacks, not a new model architecture or learning algorithm.

3 Problem Formulation

We formalise the bilateral bargaining setting instantiated by the platform and define the outcome quantities measured across all experiments.

3.1 Buyer–Seller Negotiation Setting

The platform implements a two-player alternating-offer protocol over a single commodity with a bounded round horizon, following the general structure of Rubinstein’s bargaining model [14] but with a finite horizon and hard private constraints.

In each negotiation session, one buyer and one seller exchange priced offers over at most R rounds (configurable; default $R=10$). The buyer moves first at round 0; agents alternate on subsequent rounds. On each turn, the active agent selects an action from $\{\text{OFFER}, \text{COUNTER}, \text{ACCEPT}, \text{REJECT}\}$ with an associated price (for OFFER and COUNTER) or null (for ACCEPT and REJECT).

A session terminates under one of three conditions:

- **Accepted:** an agent outputs ACCEPT, and the deal settles at the opponent’s last offered price.
- **Rejected:** an agent outputs REJECT, ending the negotiation with no deal.
- **Timeout:** the round limit R is reached without agreement.

The buyer-first turn order gives the buyer a structural first-mover advantage. Its effect on surplus distribution is documented in Section 6.

3.2 Private Constraints

Each agent holds private constraints that are never shared with the opponent and are enforced deterministically by the simulator.

Buyer. A buyer is characterised by a value v_b (maximum willingness-to-pay for the item) and a budget B_b (hard spending cap). The effective price ceiling is

$$\bar{p}_b = \min(v_b, B_b).$$

An offer at price p is feasible for the buyer if and only if $p \leq \bar{p}_b$. When $B_b \geq v_b$ (the common configuration in our experiments), the budget is slack and only the value binds.

The simulator tracks v_b and B_b separately: v_b enters the surplus calculation ($v_b - p$), while B_b enters the hard constraint check. However, the LLM agent’s prompt receives only the collapsed ceiling $\bar{p}_b$ as a single “maximum price” figure; the agent cannot reason about the value–budget distinction. Rule-based agents likewise operate on $\bar{p}_b$ directly. The separation therefore serves the simulator’s enforcement and metrics layers rather than the agent’s decision-making.

Seller. A seller is characterised by a cost c_s (reservation price, representing the minimum acceptable sale price) and a target margin m_s (aspirational markup over cost). An offer at price p is feasible for the seller if and only if $p \geq c_s$. The target margin is used only by the rule-based agent to compute its opening price ($c_s \times (1 + 2m_s)$); it is not enforced as a hard constraint and does not appear in the LLM agent’s prompt. LLM sellers receive only c_s as their “minimum price.”

Zone of possible agreement. The ZOPA [12] for a buyer–seller pair is defined as

$$\text{ZOPA} = v_b - c_s.$$

A deal is structurally possible only when $\text{ZOPA} > 0$. Budget constraints may further tighten the effective ZOPA to $\bar{p}_b - c_s = \min(v_b, B_b) - c_s$. When the ZOPA is zero or negative, no agreement can satisfy both agents’ hard constraints regardless of strategy.

3.3 Outcome Definitions

Session-level outcomes. When a session terminates with acceptance at price p , the following quantities are computed:

$$\text{buyer surplus} = v_b - p, \tag{1}$$

$$\text{seller surplus} = p - c_s, \tag{2}$$

$$\text{total welfare} = v_b - c_s. \tag{3}$$

Total welfare depends only on the buyer’s value and the seller’s cost, not on the agreed price; the price determines only the distribution of surplus between the two parties. When no deal is reached, all surplus quantities are zero. Each session additionally records the termination reason (ACCEPTED, REJECTED, or TIMEOUT) and the number of rounds taken.

Market-level outcomes. At the market level, outcomes aggregate over sessions within a tick: mean transaction price, within-tick price dispersion (standard deviation), and liquidity (fraction of matched pairs that reach a deal). Two additional metrics support mechanism-comparison experiments: *allocative efficiency*, defined as the ratio of realised total welfare to the theoretical maximum welfare across all matched pairs, and the *surplus Gini coefficient*, measuring inequality in surplus distribution across deals.

3.4 Research Questions

The seven experiments conducted on the platform operate at three levels of analysis: micro-level agent bargaining behaviour (Experiments A–C), macro-level market dynamics (Experiments D, E, I), and institutional mechanism effects (Experiment H). Each research question asks whether an observable pattern in the simulator is consistent with a specific prediction from classical theory. Specific hypotheses for each experiment are stated in Section 5.

Micro-level.

RQ-A. Do LLM agents exhibit systematic concession patterns across negotiation rounds, and how do concession trajectories differ across agent types [14]?

RQ-B. Does ZOPA width affect settlement price, and is there evidence of anchoring bias in LLM agents [5]?

RQ-C. Does varying the negotiation horizon affect settlement timing, consistent with deadline-pressure effects observed in human bargaining [13]?

Macro-level.

RQ-D. Does a market of LLM bilateral negotiators converge toward a stable price level, and does within-tick price dispersion persist [15]?

RQ-E. Do stochastic exogenous shocks produce measurable shifts in market price and liquidity in theoretically predicted directions?

RQ-I. Do structural supply-demand shifts (deterministic increases to buyer values or seller costs) produce directionally predicted price and liquidity responses?

Institutional.

RQ-H. Does the choice of buyer–seller matching algorithm affect market-level deal rates, allocative efficiency, and surplus distribution, independently of agent behaviour?

4 Simulator Architecture

The following subsections describe the platform that implements the setting formalised above: its layered architecture, agent types, validation pipeline, feasibility gate, market structure, metrics, and reproducibility mechanisms.

4.1 Overall System Architecture

The platform is organised into six layers, each with a single responsibility.

1. **Experiment Control.** CLI entry points and YAML configuration files define experimental conditions: agent types, parameter distributions, matching strategies, seed lists, and sweep grids. A dispatcher routes named experiments to dedicated runner functions that manage replication across seeds and conditions.
2. **Market Simulator.** The tick-level orchestration loop. On each tick the simulator generates buyer and seller populations from configured distributions or fixed parameter lists, applies optional demand/supply shocks (Section 4.4), delegates pair formation to a pluggable **Matcher** (Section 4.4), and dispatches each matched pair to a **NegotiationSession**. In market mode, per-tick statistics (mean price, price dispersion, liquidity) are computed and logged after all sessions in that tick complete.

3. **Negotiation Session.** A stateful object implementing the alternating-offer bargaining protocol. It maintains the transcript, the standing offer, the round counter, and termination state. The buyer always moves first (round 0); agents alternate on subsequent rounds. On each turn, the session constructs an **AgentContext** containing the agent’s private constraints, the negotiation history, and the opponent’s last offer, then passes it to the active agent’s **decide()** method. The session also hosts the feasibility gate (Section 4.3.2).
4. **Agents.** Each agent implements a **decide(ctx) → NegotiationAction** interface. Given context, agents produce a structured action (one of OFFER, COUNTER, ACCEPT, or REJECT) with an associated price, a public message visible to the opponent, and a private rationale logged for analysis but not shown to the opponent. For LLM agents, the **decide()** method internally routes the model’s raw output through a rethink loop (Section 4.3.1, Stage 2) before returning the action to the session. Five agent classes are implemented (Section 4.2).
5. **Action Judge.** A validation and enforcement layer between the agent’s raw output and the session’s state update. The judge corrects first-round protocol errors, validates offers against hard constraints (budget, cost, price bounds), and replaces invalid actions with REJECT while emitting a structured risk event (Section 4.3.1).
6. **Settlement and Metrics.** Non-LLM computation of surplus, welfare, and aggregate statistics. When a session terminates with acceptance, buyer surplus (value – price) and seller surplus (price – cost) are computed deterministically. Aggregate metrics are computed over all sessions in a run (Section 4.5).

This separation enables controlled experimentation: agent types, matching strategies, and prompt configurations can be varied independently without modifying the negotiation protocol. LLM agents internally validate their own proposed actions once via the rethink loop; external validation by the **ActionJudge** provides an independent second check. The session hosts the feasibility gate but does not construct prompts, and the simulator does not inspect negotiation transcripts.

4.2 Agent Types

All agents share a common interface: given an **AgentContext** containing the item under negotiation, the agent’s role and private constraints, the current round, the negotiation history, and the opponent’s last offer, the agent returns a **NegotiationAction**. The context structure carries value and budget separately for buyers, and cost and target margin for sellers; however, as noted in Section 3.2, LLM agent prompts present only the effective price ceiling $\bar{p}_b = \min(\text{value}, \text{budget})$ for buyers and cost for sellers.

Rule-based baseline. A deterministic agent following a linear concession strategy. Buyers open at 50% of their price ceiling and concede linearly toward it over the available rounds; sellers open at $\text{cost} \times (1 + 2m)$, where m is the target margin (default 0.15), and concede toward cost. Each round, the agent computes a concession target as

$$t_r = p_0 + (p_{\max} - p_0) \cdot \frac{r}{R - 1}$$

where p_0 is the opening price, $p_{\max}$ is the constraint boundary, r is the current round, and R is the maximum number of rounds. The agent accepts when the opponent’s standing offer meets or exceeds this target. On the final round, agents accept any offer within their hard constraints to avoid deadlock. This agent produces zero variance across seeds, providing a deterministic baseline for isolating LLM-specific effects.

LLM reactive. A single-shot LLM agent. Each prompt is assembled from modular components controlled by a `PromptConfig` object:

- A JSON-only guard and system preamble describing the alternating-offer protocol.
- The agent’s private constraints: the effective price ceiling $\bar{p}_b$ for buyers or cost floor c_s for sellers, presented as a single “maximum price” or “minimum price” figure, together with the instruction to prefer COUNTER over REJECT.
- A directional objective (e.g. “The less you pay, the better your deal”) rather than a formal surplus equation.
- Round and deadline information, with configurable deadline-salience warnings when few rounds remain.
- A decision-hint block showing the opponent’s last offer, the agent’s acceptance threshold, and the pre-computed surplus, so the model need not perform arithmetic.
- A compressed transcript of the k most recent turns (default $k=6$), formatted as terse bullet points (e.g. RO buyer OFFER \$60.00).
- The required JSON schema with two few-shot examples (one ACCEPT, one COUNTER).

The prompt explicitly instructs the model to prefer COUNTER over REJECT, on the grounds that any deal yields positive surplus while no deal yields zero. This steering reduces walk-away behaviour but does not prevent the model from rejecting if it chooses to.

The model’s raw output is processed by a rethink loop that handles two failure conditions: a JSON or schema parse error, and a parsed action that

violates the agent’s own private constraints (e.g. a buyer proposing a price above its value ceiling). On either condition, one retry is issued using the original prompt extended with a corrective message naming the specific error and a compact inline JSON template. If the retry also fails to parse, a `REJECT` action is returned, terminating the session without a deal; if the retry parses but remains infeasible, the action is passed through to the `ActionJudge` for external enforcement. The rethink loop is not configurable and applies to all LLM agent types. See Section 4.3.1 for full pipeline detail.

LLM deliberative. Identical to the reactive agent except the prompt appends a two-step reasoning scaffold that the agent must complete inside its `rationale_private` field before committing to an action: (1) assess whether the opponent’s offer is acceptable, and (2) if not, determine a counter-offer price. This scaffold is kept deliberately short for compatibility with a 3B-parameter model; longer chain-of-thought structures produced unreliable JSON output in early experiments.

This design draws on chain-of-thought prompting [17]: by requesting explicit intermediate reasoning, the prompt encourages the model to decompose the negotiation problem before acting. Whether the model performs genuine multi-step reasoning or generates post-hoc rationalisation is not verified; we measure only the observable effect on bargaining outcomes.

Memory agent. Extends the deliberative agent with episodic memory. A `MemoryStore` records the outcome of each completed negotiation (deal status, price, rounds, and an inferred opponent-style label), and the k most relevant past experiences (default $k=5$) are injected into the prompt as context. Relevance is determined by a simple heuristic: prefer interactions involving the same item, then the most recent. Memory is maintained in-memory for the duration of a single simulation run and is not persisted across runs.

Reputation agent. Further extends the memory agent with a `ReputationStore` that maintains per-opponent interaction histories. For each known opponent, the store computes a summary: number of past interactions, deal rate, average deal price, average negotiation length, and an inferred style label. Style labels are assigned by hard-coded thresholds: *cooperative* if deal rate ≥ 0.8 and average rounds ≤ 3 ; *moderate* if deal rate ≥ 0.5 ; *unreliable* if deal rate < 0.3 ; *tough* otherwise. This reputation summary is prepended to the prompt so the agent can condition its strategy on the opponent’s observed track record. The reputation agent requires per-agent memory stores (rather than role-level shared stores) and repeated pairings across ticks, provided by the round-robin matcher (Section 4.4).

4.3 Constraint Handling and Feasibility Enforcement

4.3.1 Action Validation Pipeline

Every action produced by an LLM agent passes through a five-stage pipeline before updating session state. Rule-based agents skip Stages 0 and 2, as they produce deterministic, constraint-respecting actions by construction.

Stage 0: Pre-LLM settlement gate. Before calling the agent’s `decide()` method, the session checks whether the opponent’s standing offer is already feasible for the current agent. If so, the session settles immediately without invoking the LLM. This mechanism is described in full in Section 4.3.2; it applies only when `gate_enabled` is `true`, which is the configuration used in all seven reported experiments.

Stage 1: JSON extraction and schema validation. For LLM agents, the raw text output must be parsed into a structured action. The parser attempts four extraction strategies in sequence: (i) direct JSON parsing, (ii) extraction from markdown code fences, (iii) brace-delimited substring extraction, and (iv) heuristic repair (single-to-double quote conversion, trailing comma removal, unquoted key correction). The extracted object is checked against the action schema: it must contain an `action` field with a value in `{OFFER, COUNTER, ACCEPT, REJECT}`, an `offer_price` (required and positive for `OFFER/COUNTER`, coerced to null for `ACCEPT/REJECT`), and a `message_public` string.

Stage 2: Rethink loop. If Stage 1 fails (the output cannot be parsed into a valid action), or if the parsed action violates the agent’s own private constraints, one rethink attempt is issued. The constraint check applies the following tests: buyer offers and counters must not exceed $\min(\text{value}, \text{budget})$ and must have a positive price; seller offers and counters must not fall below cost and must have a positive price; an `ACCEPT` action requires a standing offer, and the standing offer must itself satisfy the accepting agent’s constraint. These checks do not cover global price bounds $[p_{\min}, p_{\max}]$, which are enforced only by the `ActionJudge` in Stage 3. On triggering, the original prompt is extended with a corrective message naming the specific error and a compact inline JSON template showing the four required fields; the extended prompt is submitted as a single new inference call.

Recovery follows two paths depending on the rethink outcome. **(a)** If the rethink response parses successfully, it is returned as the agent’s action regardless of feasibility; any remaining constraint violation is logged at warning level but not enforced by the rethink loop, and is left for Stage 3 to handle. **(b)** If the rethink response also fails to parse: the first-attempt action is returned if it was parseable (even though infeasible), or a `REJECT` action is returned if the first attempt was entirely unparseable, terminating the session without a deal.

The rethink loop is not configurable and is always active for all LLM agent types. It is distinct in purpose from the feasibility gate (Stage 0): rethink

corrects output errors in the model’s own response (format or self-constraint violations), while the gate corrects decision errors (failure to accept an offer that would yield positive surplus). Rethink events are logged individually but their frequency is not separately aggregated in the published experimental results.

Stage 3: Protocol and constraint validation. The `ActionJudge` applies domain-level checks. First-round corrections handle protocol violations: a `COUNTER` on round 0 is relabelled as `OFFER` with the proposed price preserved; an `ACCEPT` or `REJECT` on round 0 is replaced with an `OFFER` at the agent’s proposed price if available, otherwise at a default opening price (50% of value for buyers, $1.5 \times$ cost for sellers). Constraint validation checks hard limits: buyer offers must not exceed budget or value; seller offers must not fall below cost; all prices must lie within configurable global bounds $[p_{\min}, p_{\max}]$.

Stage 4: Enforcement. If validation fails, the judge replaces the action with `REJECT` and emits a structured risk event recording the violation type (budget, cost, bounds, or logic), the attempted action and price, the round number, and the time step. Risk events are logged in the JSONL event stream and aggregated in the run summary, enabling post-hoc analysis of constraint-violation frequency across agent types and conditions.

4.3.2 Feasibility Gate

A configurable deterministic settlement mechanism that can override agent decisions at two points within each negotiation turn. Both components are controlled by a single boolean flag (`gate_enabled`, default `true`); when disabled, neither fires.

Mechanism. Before calling the agent’s `decide()` method, the session evaluates whether the opponent’s standing offer is feasible for the current agent. Feasibility requires non-negative utility: for a buyer, the offer must satisfy $\text{offer} \leq \min(\text{value}, \text{budget})$; for a seller, $\text{offer} \geq \text{cost}$ (cf. [12]). If feasible, the session settles immediately. This addresses a practical failure mode: small language models frequently fail to output `ACCEPT` even when the offer is clearly profitable, causing deadlock when a zone of possible agreement unambiguously exists.

A secondary post-LLM check applies the same feasibility test after the agent acts, overriding `COUNTER` or `REJECT` to `ACCEPT` if the standing offer is feasible. Both gate components log the LLM’s intended action alongside the effective (possibly overridden) action, an override flag, and the computed utility, allowing post-hoc separation of LLM-generated decisions from gate-driven settlements.

Trade-off. The gate ensures high deal rates in wide-ZOPA scenarios but masks behavioural effects requiring multi-round interaction, notably deadline

pressure and extended concession trajectories. Experiments targeting these effects should disable the gate and accept lower deal rates as part of the measurement. We report gate-enabled results alongside the gate’s override rate so readers can assess whether settlement outcomes are agent-driven or mechanically enforced. The implications of this trade-off for experimental interpretation are discussed in Section 7.

4.4 Market Layer

Population generation. The simulator supports two scenario modes. In *distribution mode*, buyer values, budgets, seller costs, and target margins are drawn independently from uniform distributions with configurable bounds (e.g. buyer value $\in$ [\$80, \$150], seller cost $\in$ [\$50, \$120]). In *fixed mode*, all agents receive identical parameter values specified in the configuration, enabling controlled single-variable experiments. In both modes, populations are regenerated each tick from a forked RNG, so agents within a tick are independent draws.

Matching strategies. Pair formation is delegated to a pluggable *Matcher* interface. Four implementations are provided:

- **Random** (default). Shuffles buyers and sellers independently and pairs them 1:1. Unmatched agents (when population sizes differ) are dropped. This is the baseline matching mechanism used in experiments A–E.
- **Surplus-maximising** (oracle). Computes $ZOPA = \text{buyer value} - \text{seller cost}$ for all possible pairs, then greedily assigns pairs in descending ZOPA order (each agent used at most once). Only pairs with positive ZOPA are matched. This uses private information and represents a theoretical upper bound on matching efficiency, not a realistic mechanism.
- **Sorted** (heuristic). Sorts buyers by value descending and sellers by cost descending, then pairs them positionally. This tends to create moderate but consistent ZOPAs without requiring private information beyond the ordering.
- **Round-robin** (repeated). On the first tick, pairs agents 1:1 and remembers the pairings. On subsequent ticks, agents with the same IDs are re-paired, enabling reputation and memory effects to accumulate across interactions. New agents are paired randomly with unmatched partners.

Demand and supply shocks. An optional shock mechanism applies multiplicative perturbations to buyer values and seller costs at the start of each tick. Shocks fire with a configurable probability (default 10%) per tick. When a shock fires, buyer values are scaled by a demand multiplier drawn uniformly from a configured range (e.g. [0.7, 1.3]) and seller costs are scaled by an independently drawn supply multiplier from the same range. Budgets are not modified by shocks, preserving hard spending constraints. This mechanism enables experiments on market resilience and price response to exogenous perturbations.

4.5 Metrics and Logging

Session-level metrics. Each completed negotiation produces a structured result containing: deal status, agreed price (if any), number of rounds, termination reason, buyer and seller surplus, and a list of risk events.

Run-level aggregates. After all sessions in a run complete, the following metrics are computed:

- **Deal success rate:** fraction of sessions resulting in a deal.
- **Price statistics:** mean, median, and standard deviation of deal prices.
- **Surplus:** mean buyer surplus, mean seller surplus, and mean total welfare.
- **Rounds:** mean rounds to close (deals only) and mean rounds across all sessions.
- **Deadlock rate:** fraction of sessions terminating by timeout.
- **Risk events:** counts of budget violations, cost violations, and total constraint violations.

Tick-level statistics (market mode). In market mode, the simulator computes per-tick aggregates after each tick’s sessions complete: number of sessions, deals made, fail rate, mean transaction price, within-tick price standard deviation, liquidity (deals / sessions), and mean buyer and seller surplus. These are logged as `tick_end` events and enable time-series analysis of market dynamics.

Efficiency and inequality. Two additional metrics support mechanism-comparison experiments. *Allocative efficiency* is the ratio of realised total welfare to the theoretical maximum welfare (computed as $\sum \max(\text{value}_i - \text{cost}_i, 0)$ over all matched pairs, where unrealised deals contribute zero welfare). The *surplus Gini coefficient* measures inequality in surplus distribution across deals, with 0 indicating perfect equality and 1 indicating maximum concentration.

Structured event logging. Every negotiation action, session outcome, constraint violation, and tick-level aggregate is written to a newline-delimited JSON log (`events.jsonl`). Turn events include the round number, role, action type, offer price, and public message. For LLM agents, the log additionally records the intended action (what the model output), the effective action (what the simulator executed), an override flag, the computed utility, and the feasibility threshold. This granular logging supports post-hoc analysis—including separation of LLM-driven and gate-driven settlements—without re-execution.

Reproducibility. All stochastic components—buyer/seller parameter generation, item assignment, pair matching, shock application—draw from a `SeededRNG` wrapper around Python’s `random.Random`. At each tick, the simulator calls `fork()`, which consumes one value from the parent RNG to derive a child seed, producing an independent child generator for that tick. LLM inference introduces non-determinism from the model itself, but all market-level processes remain fully reproducible given the same seed. All experimental parameters are specified in YAML files with CLI override support, and each experiment runner records the active configuration, the git commit hash, and the seed list in a structured JSON summary.

5 Methodology and Experimental Design

This section describes the methodology and experimental design for all seven experiments conducted on the platform. The experiments are organised into three levels of analysis—micro-level agent bargaining behaviour (Experiments A–C), macro-level market outcomes (Experiments D, E, I), and institutional mechanism effects (Experiment H)—reflecting the platform’s ability to connect individual negotiation dynamics to aggregate market properties within a single system. Results are reported in Section 6.

5.1 Experimental Philosophy

The experimental programme is designed around three principles.

First, *controlled comparison against theory*. Each experiment targets a specific prediction from classical bargaining or market theory and varies a single independent variable while holding all other conditions constant. This enables directional comparison against theoretical baselines, though the small number of seeds and the absence of formal statistical tests limit findings to descriptive characterisation rather than confirmatory inference.

Second, *micro-to-macro connection*. The platform supports experiments at both the individual session level (where agent behaviour can be observed turn-by-turn) and the market level (where aggregate statistics emerge from many sessions per tick). This multi-level design enables investigation of whether micro-level bargaining patterns translate into macro-level market regularities.

Third, *transparent confound reporting*. Two design choices — the feasibility gate and prompt steering — are active throughout most experiments and affect results in ways that cannot be separated from agent behaviour without ablation. Rather than treating these as neutral background conditions, we report their involvement rates per experiment and discuss their implications for interpretation. The gate and prompt architecture are treated as experimental confounds, not as controlled variables, in the current study; ablation is identified as a priority for future work.

5.2 Common Setup

All experiments share the following configuration unless explicitly overridden.

Model and inference. A single language model is used throughout: `llama3.2:3b` served locally via Ollama at temperature 0.2, with a maximum output length of 256 tokens and a 30-second timeout per call. Failed LLM calls are retried once with a format-correction prompt (the rethink loop described in Section 4.3.1). If the retry also fails to parse, a `REJECT` action is returned, terminating the session without a deal; if the retry parses but remains infeasible, the action is passed through to the `ActionJudge` for enforcement.

Feasibility gate. The deterministic settlement gate (Section 4.3.2) is enabled for all seven experiments. As discussed in Section 4.3.2, this ensures high deal rates in wide-ZOPA scenarios but masks behavioural effects that require multi-round interaction. The gate’s override rate is reported per experiment in Section 6 so that gate-driven and LLM-driven settlements can be distinguished.

Prompt steering. Two prompt-level design choices apply across all LLM conditions and affect interpretation:

- The prompt explicitly instructs the model to prefer `COUNTER` over `REJECT`, reducing walk-away behaviour. This steering is active in all experiments and may suppress deadline effects (Experiment C) and inflate deal rates relative to an unsteered baseline.
- An accept-condition block pre-computes the utility of the opponent’s standing offer and presents it directly in the prompt, so the model need not perform arithmetic. This may reduce anchoring effects (Experiment B) by giving the model explicit surplus information rather than requiring it to infer value from the first offer.

Neither of these design choices has been ablated; their effects on results are discussed as confounds rather than controlled variables.

Reproducibility. All stochastic processes are controlled by a `SeededRNG` with per-tick forking (Section 4.5). Each experiment records the active configuration, seed list, and git commit hash in a structured JSON summary. Every agent action, session outcome, and constraint violation is logged to a JSONL event stream.

Replication. Experiments A–C, H, and I are each replicated across three random seeds (42, 123, 456). Experiments D and E use a single seed (42) and are reported as exploratory. Table 1 summarises all seven experiments.

Table 1: Summary of all experiments, organised by level of analysis.

Level	Exp	Name	Independent variable	Agent type	Sessions	Seeds
Micro	A	Concession	agent type (3 levels)	mixed	450	3
	B	Anchoring	ZOPA width (3 levels)	reactive	450	3
	C	Deadline	max rounds (3 levels)	deliberative	450	3
Macro	D	Market	time (30 ticks)	reactive	450	1
	E	Shock (stochastic)	shock on/off	reactive	900	1
	I	Shock (structural)	supply/demand shift (3 levels)	reactive	900	3
Institutional	H	Mechanism	matching algorithm (2 levels)	reactive	520	3
Total					4,120	

5.3 Micro-Level Experiments: Agent Bargaining Behaviour

Experiments A–C use fixed-scenario parameters to isolate individual agent behaviour from market-level variation. Each experiment tests whether LLM agents reproduce a specific prediction from classical bargaining theory: systematic concession dynamics [14], anchoring bias [5], and deadline-driven concession acceleration [13].

5.3.1 Experiment A: Concession Dynamics

Research questions.

- RQ-A1.** Do LLM agents exhibit systematic concession patterns across negotiation rounds?
- RQ-A2.** How do concession trajectories differ between rule-based, reactive, and deliberative agents?
- RQ-A3.** Does chain-of-thought prompting (deliberative) alter concession behaviour relative to single-shot prompting (reactive)?

Independent variable. Agent type, varied across three conditions: `rule_based`, `llm_reactive`, and `llm_deliberative`. Both buyer and seller use the same agent type within each condition.

Controlled variables. All sessions use fixed-mode parameters to isolate the effect of agent type from scenario variation. The fixed scenario is: buyer value = \$120, seller cost = \$80, buyer budget = \$130, item reference price = \$100, seller target margin = 15%. This creates a ZOPA of \$40 (\$120 − \$80), wide enough for reliable settlement. Maximum rounds = 10. Feasibility gate is enabled.

Design. 3 agent-type conditions × 3 seeds × 50 sessions per seed = 450 sessions total.

Metrics.

- Deal rate (%) per condition.
- Mean agreed price (\$ $\pm$ standard deviation across seeds).
- Mean rounds to settlement.
- Per-round offer price trajectory (concession curve), plotted per agent type and role.
- Buyer and seller surplus.
- Gate involvement rate: fraction of acceptances driven by the feasibility gate rather than agent decision.

Hypotheses.

- H-A1.** All three agent types achieve $>90\%$ deal rate given the wide ZOPA.
- H-A2.** LLM agents settle in fewer rounds than the rule-based agent, whose linear concession formula spreads concessions across the full horizon.
- H-A3.** The deliberative agent produces different counter-offer prices than the reactive agent, reflecting the effect of structured reasoning on offer generation.

5.3.2 Experiment B: Anchoring Effects

Research questions.

- RQ-B1.** Does ZOPA width—manipulated via buyer value—affect settlement price?
- RQ-B2.** Do LLM agents exhibit anchoring bias, where the first offer pulls the final price toward it?

Independent variable. Buyer value, varied across three levels: \$100, \$120, and \$150. Seller cost is fixed at \$80, giving ZOPA widths of \$20, \$40, and \$70 respectively. Buyer budget is set \$10 above buyer value in each condition (\$110, \$130, \$160) to avoid budget-binding confounds.

Controlled variables. Agent type is `llm_reactive` in all conditions. Seller target margin = 15%, item reference price = \$100, maximum rounds = 10, feasibility gate enabled.

Design. 3 ZOPA conditions $\times$ 3 seeds $\times$ 50 sessions per seed = 450 sessions total.

Metrics.

- Deal rate per ZOPA level.
- Mean settlement price per ZOPA level ($\$ \pm$ standard deviation).
- Mean first offer (buyer’s round-0 price) per condition.
- Pearson correlation r between first-offer price and final deal price, computed across all deals.
- Buyer surplus per condition.

Hypotheses.

- H-B1.** Wider ZOPA leads to higher deal rates. Narrow ZOPA (\$20) may produce deal failure if the seller’s opening counter exceeds the buyer’s hard cap.
- H-B2.** If anchoring bias is present, the first-offer/final-price correlation r should be positive and significant: higher first offers should pull settlement prices upward.
- H-B2’.** If the accept-condition block overrides anchoring, settlement prices may be insensitive to first-offer level, yielding $r \approx 0$.

5.3.3 Experiment C: Deadline Pressure

Research questions.

- RQ-C1.** Does varying the negotiation horizon (maximum rounds) affect when deals settle?
- RQ-C2.** Do LLM agents exhibit deadline effects—accelerated concession near the final rounds—consistent with the findings of Roth et al. [13]?

Independent variable. Maximum rounds per session, varied across three levels: 4, 8, and 16.

Controlled variables. Agent type is `llm_deliberative` in all conditions, chosen because the deliberative agent’s structured reasoning offers the best opportunity for deadline-aware strategy adjustment. Fixed scenario parameters match Experiment A (buyer value = \$120, seller cost = \$80, ZOPA = \$40). Feasibility gate is enabled. The prompt includes configurable deadline-salience language that warns the agent when few rounds remain.

Design. 3 horizon conditions $\times$ 3 seeds $\times$ 50 sessions per seed = 450 sessions total.

Metrics.

- Deal rate per horizon condition.
- Mean settlement round per condition (mean $\pm$ standard deviation).
- Settlement round as a fraction of maximum rounds (normalised timing).
- Share of agreements occurring in the last two rounds of the available horizon.

Hypotheses.

- H-C1.** If agents are deadline-sensitive, longer horizons should produce later settlements (agents use available time) and a disproportionate share of agreements near the deadline.
- H-C1'.** If the feasibility gate dominates settlement timing, the settlement round should be constant regardless of horizon, as the gate fires whenever the standing offer becomes feasible—typically in the first few rounds.

Known confounds. Two design choices work against observing deadline effects. First, the feasibility gate settles deals as soon as offers become feasible, typically before deadline proximity is relevant. Second, the prompt steers agents to prefer COUNTER over REJECT, removing the primary mechanism by which agents could delay settlement. These confounds are not ablated in this experiment; their implications are discussed in Section 6.

5.4 Macro-Level Experiments: Market Outcomes

Experiments D, E, and I shift the unit of analysis from individual sessions to multi-tick markets populated by heterogeneous LLM agents. These experiments test whether aggregate market outcomes—price equilibrium, liquidity, and shock response—are consistent with predictions from bilateral bargaining theory [15] and supply-demand theory. All three use distribution-mode parameters, random matching, and reactive agents.

5.4.1 Experiment D: Market Price Discovery

Research questions.

- RQ-D1.** Does a multi-tick market populated by LLM bilateral negotiators converge toward a stable price level?
- RQ-D2.** Does price dispersion persist within ticks, consistent with the bilateral bargaining prediction that the Law of One Price fails when pairs negotiate independently [15]?

Independent variable. Time, observed longitudinally over 30 market ticks. There is no between-condition manipulation; this is a single-condition observational study.

Controlled variables. The simulator runs in market mode with 15 buyer-seller pairs per tick, using random matching. Agent type is `llm_reactive`. Scenario mode is *distribution*: buyer values are drawn uniformly from [\$80, \$150], seller costs from [\$50, \$120], buyer budgets from [\$100, \$200], and seller margins from [5%, 25%]. Maximum rounds = 10. Feasibility gate is enabled.

Design. 30 ticks $\times$ 15 sessions per tick = 450 sessions, single seed (42). This is an exploratory design; results should be interpreted with the single-seed caveat.

Metrics.

- Mean transaction price per tick, with within-tick standard deviation.
- Liquidity per tick: fraction of matched pairs that reach a deal.
- Price trend: slope of a simple linear regression of mean price over ticks.
- Mean buyer surplus and mean seller surplus per tick.
- Early-vs-late comparison: mean price and liquidity in ticks 0–4 versus ticks 25–29.

Hypotheses.

- H-D1.** Mean price stabilises within the first 10 ticks, with a near-zero price trend slope thereafter.
- H-D2.** Within-tick price dispersion remains substantial (standard deviation $>$ \$10), consistent with bilateral bargaining theory.
- H-D3.** Buyer surplus exceeds seller surplus on average, reflecting the buyer’s first-mover advantage in the alternating-offer protocol.

5.4.2 Experiment E: Stochastic Shock Response

Research questions.

- RQ-E1.** Do exogenous demand and supply shocks produce measurable shifts in market price and liquidity?
- RQ-E2.** Is the direction of price adjustment consistent with economic theory (demand increase $\rightarrow$ price increase; supply increase $\rightarrow$ price decrease)?

Independent variable. Shock condition, varied across two levels: `no_shock` (baseline) and `with_shock`. In the shock condition, each tick has a 30% probability of a shock event. When a shock fires, buyer values are scaled by a demand multiplier drawn uniformly from $[0.7, 1.3]$ and seller costs are scaled by an independently drawn supply multiplier from the same range. Budgets are not modified.

Controlled variables. Both conditions use the same market-mode configuration as Experiment D: 30 ticks, 15 pairs per tick, random matching, distribution-mode parameters, `llm_reactive` agents, feasibility gate enabled.

Design. 2 conditions $\times$ 30 ticks $\times$ 15 sessions per tick = 900 sessions total, single seed (42). Exploratory design.

Metrics.

- Mean transaction price per tick, compared across conditions.
- Within-tick price standard deviation per condition.
- Mean liquidity per condition, with minimum and maximum tick-level values.
- Price and liquidity differences (Δ) between shock and no-shock conditions.

Hypotheses.

- H-E1.** The shock condition produces higher price volatility (larger tick-to-tick price variance) than the baseline.
- H-E2.** Mean price and mean liquidity are lower under shocks, because negative shocks (demand multiplier < 1 or supply multiplier > 1) compress the ZOPA for affected pairs.
- H-E3.** Worst-case liquidity drops more severely under shocks than best-case liquidity improves, producing an asymmetric tail risk.

5.4.3 Experiment I: Structural Supply-Demand Shifts

Experiment E applies stochastic multiplicative shocks of unknown direction. Experiment I complements this with deterministic, directional shifts to buyer values or seller costs, enabling cleaner tests of whether LLM-agent markets respond to supply and demand conditions in theoretically predicted directions.

Research questions.

- RQ-I1.** Do transaction prices shift in the direction predicted by supply-demand theory when buyer values increase (demand shift) or seller costs increase (supply shift)?
- RQ-I2.** Does the magnitude of price and liquidity response differ between demand-side and supply-side shifts?
- RQ-I3.** Can observed liquidity changes be decomposed into mechanical effects (fewer feasible pairs) and behavioural effects (lower deal success among feasible pairs)?

Independent variable. Supply-demand condition, varied across three levels:

- **baseline:** buyer values drawn from [\$80, \$150], seller costs from [\$50, \$120] (same ranges as Experiments D and E).
- **demand_shock:** buyer values shifted upward by \$20, drawn from [\$100, \$170]; seller costs unchanged.
- **supply_shock:** seller costs shifted upward by \$20, drawn from [\$70, \$140]; buyer values unchanged.

Controlled variables. The simulator runs in market mode with 10 buyer-seller pairs per tick over 10 ticks. Agent type is `llm_reactive`. Random matching. Maximum rounds = 10. Feasibility gate is enabled.

Design. 3 conditions $\times$ 3 seeds (42, 123, 456) $\times$ 10 ticks $\times$ 10 pairs per tick = 900 sessions total (300 per condition).

Metrics.

- Mean transaction price per condition.
- Liquidity (deal rate) per condition.
- Total welfare, mean buyer surplus, and mean seller surplus per condition.
- Within-condition price dispersion (standard deviation).
- Feasibility decomposition: fraction of pairs with positive ZOPA (mechanical), deal success rate among feasible pairs (behavioural), and overall liquidity.

Hypotheses.

- H-I1.** A demand shift (higher buyer values) increases mean transaction price and liquidity: more pairs have positive ZOPA, and buyers with higher values accept higher prices.
- H-I2.** A supply shift (higher seller costs) increases mean transaction price but decreases liquidity: sellers demand higher prices to cover costs, and fewer pairs have sufficient ZOPA for agreement.
- H-I3.** Liquidity effects are partly mechanical (ZOPA compression) and partly behavioural (LLM agents are less effective at reaching agreement when the ZOPA is narrow).

5.5 Institutional Experiment: Mechanism Design

Experiment H tests whether an institutional design choice—the buyer–seller matching algorithm—affects market-level outcomes independently of agent behaviour. This complements the micro-level and macro-level experiments by examining whether market structure can be improved through mechanism design even when agent sophistication is held constant.

5.5.1 Experiment H: Matching Mechanism Comparison

Research question. Does the matching algorithm affect market-level deal rates, allocative efficiency, welfare, and price dispersion?

Independent variable. Matching algorithm, varied across two conditions: **random** (baseline, all agents paired regardless of ZOPA) and **surplus_max** (oracle, greedy maximum-ZOPA pairing that only matches pairs with positive ZOPA overlap). The platform additionally implements sorted and round-robin matchers, whose empirical evaluation is deferred to future work.

Controlled variables. The simulator runs in market mode with 10 buyer–seller pairs per tick over 10 ticks. Agent type is **llm_reactive**. Scenario mode is *distribution*, using the same parameter ranges as Experiments D and E (buyer values from [\$80, \$150], seller costs from [\$50, \$120]). Maximum rounds = 10. Feasibility gate is enabled.

Design. 2 matching conditions $\times$ 3 seeds (42, 123, 456) $\times$ 10 ticks $\times$ 10 pairs per tick = 520 sessions total (300 random + 220 surplus-max; the surplus-max condition produces fewer sessions because it excludes negative-ZOPA pairs).

Metrics.

- Deal rate per condition.

- Mean transaction price and price standard deviation per condition.
- Allocative efficiency: realised welfare as a fraction of theoretically maximum welfare.
- Surplus Gini coefficient: inequality of surplus distribution across deals.
- Mean buyer surplus and mean seller surplus per condition.

Hypotheses.

- H-H1.** Surplus-maximising matching produces a higher deal rate than random matching, because it eliminates structurally infeasible pairs.
- H-H2.** Surplus-maximising matching produces higher allocative efficiency, because it concentrates trades in high-surplus pairs.

5.6 Deferred Experiments

The platform supports two additional experiment families for which infrastructure is implemented but empirical evaluation has not been conducted. Experiment F (Reputation and Repeated Interaction) uses the round-robin matcher and reputation agent to test whether access to opponent interaction history improves deal rates over time. Experiment G (Communication Strategy) uses prompt-level communication modifiers (neutral, assertive, collaborative, strategic) to test whether message framing affects concession speed and settlement price. Both experiments require the feasibility gate to be disabled so that strategic adaptation can manifest over multiple rounds. Their designs and evaluation are deferred to future work (Section 8).

5.7 Controls and Confounds

Several design choices function as uncontrolled confounds across the experimental programme. We enumerate them here so that their effects can be tracked when interpreting results in Section 6.

Feasibility gate. The deterministic settlement gate (Section 4.3.2) is enabled in all seven experiments. In wide-ZOPA fixed-scenario experiments (A–C), the gate fires at round 2 in every session, limiting observed concession data to two rounds and making settlement timing invariant to horizon manipulation. In market-mode experiments (D, E, H, I), gate involvement varies per session depending on the drawn ZOPA. The gate’s contribution to deal rates and settlement timing cannot be separated from agent capability without ablation.

Prompt steering. Two prompt-level choices are active in all LLM conditions: the prefer-COUNTER instruction (which suppresses rejection and may inflate deal rates) and the accept-condition block (which pre-computes utility and may reduce anchoring effects). Neither has been ablated; their separate effects are unknown.

Single model. All experiments use `llama3.2:3b` at temperature 0.2. Results may not generalise to other model families, scales, or temperatures.

Seed limitations. Experiments D and E use a single seed (42) and are reported as exploratory. Experiments A–C, H, and I use three seeds, supporting directional replication but not formal statistical testing.

5.8 Evaluation Protocol

For each experiment, we report the following:

1. **Primary outcome metrics** as specified in the experiment design (deal rate, settlement price, surplus, timing, or market-level aggregates).
2. **Gate involvement rate:** the fraction of acceptances driven by the feasibility gate rather than by agent decision, enabling readers to assess the degree of mechanical versus behavioural settlement.
3. **Hypothesis evaluation:** whether each stated hypothesis is supported, rejected, or indeterminate, using descriptive comparison (means, standard deviations, Pearson correlation) rather than formal statistical tests.
4. **Confound acknowledgement:** for each finding, whether the result is attributable to agent behaviour, protocol structure, gate enforcement, or some combination that the current design cannot separate.

All analyses are descriptive. No hypothesis tests, confidence intervals, or effect sizes are computed. Findings are characterised as “consistent with,” “directionally aligned with,” or “inconsistent with” theoretical predictions, rather than as confirmations or rejections in the statistical sense.

6 Results

This section reports results for the seven completed experiments (A–E, H, I) described in Section 5. We begin with sanity checks on the LLM backend and gate involvement, then present micro-level results (Experiments A–C), macro-level results (Experiments D, E, I, H), and conclude with a cross-experiment synthesis.

6.1 Sanity Checks and Agent Validity

Before examining experimental outcomes, we report the reliability of the LLM backend and the feasibility gate’s involvement across all 4,120 sessions.

Table 2: LLM backend quality and feasibility gate involvement across all experiments.

Metric	Value
LLM fallback rate (format failures)	<0.1%*
Post-LLM override rate	0.0%
Gate-driven acceptances (fraction of all turns)	~40%
LLM-driven decisions (fraction of all turns)	~60%

The `11ama3.2:3b` model produced valid JSON on nearly every call (*Experiment E produced 2 parse failures and 1 timeout across 900 sessions; the retry mechanism handled all three without data loss). The post-LLM feasibility check (which overrides COUNTER or REJECT to ACCEPT when the standing offer is feasible) was never triggered, indicating that when the gate did not pre-empt the LLM call, the model never rejected a clearly profitable offer.

The gate drove approximately 40% of all acceptances. In fixed-scenario experiments (A–C), the gate fires deterministically after the seller’s round-1 counter-offer whenever that counter is within the buyer’s constraints, settling all sessions at round 2. In market-mode experiments (D, E, I), gate involvement varies per session depending on the randomly drawn ZOPA for each pair.

6.2 Micro-Level Results: Agent Bargaining Behaviour

Experiments A–C use fixed-scenario parameters to isolate individual agent behaviour. The feasibility gate is enabled in all three experiments and drives 100% of acceptances, as documented in Section 6.1. All findings in this subsection must therefore be interpreted as characterising the combined system (LLM + gate + prompt), not LLM decision-making in isolation.

6.2.1 Experiment A: Concession Dynamics

Table 3 reports the mean offer price at each observed round for the three agent-type conditions. All 450 sessions (150 per condition) resulted in deals, consistent with H-A1; however, all settlements were gate-driven (the gate fired at round 2 in every session), so the 100% deal rate reflects gate enforcement rather than agent negotiation capability.

All sessions settled at round 2 via the feasibility gate, which fired after the seller’s round-1 counter-offer. This limits the concession data to two observations per session.

Table 3: Mean offer price by agent type and round ($n = 150$ per condition, pooled across 3 seeds). Concession Δ is the absolute difference between the seller’s round-1 counter and the buyer’s round-0 opening.

Condition	Round 0 (Buyer opens)	Round 1 (Seller counters)	Concession Δ
<code>rule_based</code>	$\$60.00 \pm \0.00	$\$101.33 \pm \0.00	$\$41.33$
<code>llm_reactive</code>	$\$65.46 \pm \0.90	$\$103.78 \pm \3.05	$\$38.32$
<code>llm_deliberative</code>	$\$66.74 \pm \0.67	$\$96.57 \pm \2.72	$\$29.83$

Finding 1: LLM agents open less aggressively than the rule-based baseline. Both LLM conditions open \$5–7 higher than the rule-based agent (\$65–67 vs. \$60). The rule-based buyer opens at exactly 50% of its value ceiling, while the LLM agents do not reproduce this heuristic precisely.

Finding 2: Chain-of-thought prompting produces more conservative counter-offers. The deliberative agent’s seller counter (\$96.57) is \$7.21 lower than the reactive agent’s (\$103.78), yielding a concession gap of \$29.83 versus \$38.32. This is a prompting effect—the two-step reasoning scaffold (“Is the offer acceptable? If not, what price?”) leads the model to hold a firmer position—rather than evidence of Rubinstein-style time-preference patience. H-A3 is supported: structured reasoning alters counter-offer behaviour.

Finding 3: Rule-based agent produces zero variance. Standard deviation of \$0.00 across all 150 sessions per round confirms that the seeded RNG and linear concession formula are fully deterministic, validating the platform’s reproducibility guarantees.

Limitation. The feasibility gate forces settlement at round 2 in all sessions, meaning only two data points per session are observed. A full concession trajectory over 10 rounds is not available in this experiment. Future work should test with the gate disabled or a narrower ZOPA to observe multi-round concession curves.

6.2.2 Experiment B: Anchoring Effects

Table 4 reports deal rates, first offers, and settlement prices across the three ZOPA conditions.

First-offer / final-price Pearson $r = -0.148$ across all deals.

Finding 4: ZOPA width is the primary determinant of deal success. The `anchor_low` condition collapses to a 2.7% deal rate compared to 100% in the other two conditions. With $ZOPA = \$20$, the seller’s opening counter of approximately \$104 exceeds the buyer’s hard cap of \$100, making the offer infeasible. The feasibility gate cannot fire, and the LLM must negotiate the

Table 4: Anchoring experiment outcomes by condition ($n = 150$ per condition, pooled across 3 seeds). Seller cost is fixed at \$80 in all conditions.

Condition	Deal rate	First offer	Final price	Buyer surplus
<code>anchor_low</code> (ZOPA = \$20)	2.7% (4/150)	\$55.00 $\pm$ \$0.00	\$96.40 $\pm$ \$0.00	\$3.60
<code>anchor_mid</code> (ZOPA = \$40)	100% (150/150)	\$65.51 $\pm$ \$1.03	\$103.48 $\pm$ \$3.25	\$16.52
<code>anchor_high</code> (ZOPA = \$70)	100% (150/150)	\$75.00 $\pm$ \$0.00	\$101.55 $\pm$ \$5.50	\$48.45

seller below its natural opening position—a task it nearly always fails. H-B1 is supported.

Finding 5: Classical anchoring bias is absent (null result). Pearson $r = -0.148$ indicates no meaningful positive relationship between first offer and final price. The `anchor_high` condition (first offer \$75) does not produce higher settlement prices than `anchor_mid` (first offer \$65); final prices converge to the \$101–103 range regardless. H-B2 is rejected; H-B2' is supported.

This is a departure from human bargaining results, where first offers reliably anchor final outcomes [5]. One possible explanation is that the accept-condition block in the prompt provides explicit surplus information, reducing the model’s reliance on the first offer as an informational signal. This hypothesis has not been tested via ablation.

Design confound. The independent variable simultaneously manipulates ZOPA width and first-offer level, making it impossible to fully isolate anchoring from ZOPA feasibility effects. The Pearson r metric partially addresses this by measuring within-deal correlation, but the confound remains.

6.2.3 Experiment C: Deadline Pressure

Table 5 reports settlement timing across the three horizon conditions.

Table 5: Deadline experiment outcomes by time horizon ($n = 150$ per condition, pooled across 3 seeds).

Max rounds	Deal rate	Avg closing round	SD	Last-2-round share
4	100%	3.00	0.00	100.0%
8	100%	3.00	0.00	0.0%
16	100%	3.00	0.00	0.0%

Finding 6: Deadline horizon does not affect settlement timing (null result). All conditions close at exactly round 3 with zero variance, regardless of whether the time limit is 4, 8, or 16 rounds. The feasibility gate fires deterministically after the seller’s round-1 counter-offer, producing a mechanical

settlement at round 2 that is fully independent of deadline proximity. H-C1 is not supported; the outcome is consistent with H-C1'.

Finding 7: The `max_rounds=4` last-2-round share is an arithmetic artifact. The 100% last-2-round share under `max_rounds=4` occurs because round 3 of 4 falls within the “last 2 rounds” window. This is not evidence of deadline-driven concession; it is a consequence of the gate’s fixed firing time relative to a short horizon.

Confounds. Two design choices compound to produce this null result. First, the feasibility gate settles deals before deadline proximity becomes relevant. Second, the prompt steers agents to prefer COUNTER over REJECT, removing the primary mechanism by which agents could delay settlement. To study genuine deadline effects, future work should disable the gate and remove the prefer-counter instruction, accepting lower deal rates as part of the measurement.

6.3 Macro-Level Results: Market Outcomes and Mechanism Effects

Experiments D, E, I, and H shift the unit of analysis from individual sessions to multi-tick markets populated by heterogeneous agents. Gate involvement varies per session depending on the drawn ZOPA. Experiments D and E use a single seed and are reported as exploratory; Experiments H and I are replicated across three seeds.

6.3.1 Experiment D: Market Price Discovery

Tables 6 and 7 report aggregate market statistics and temporal comparison for the 30-tick market simulation.

Table 6: Market dynamics aggregate statistics (30 ticks, single seed).

Metric	Value
Mean transaction price	\$89.44
Price range	\$78.12 – \$110.03
Within-tick price std (mean)	\$20.02
Average liquidity	52.2%
Liquidity range	13.3% – 80.0%
Price trend slope	+\$0.057 / tick
Buyer surplus (mean)	\$32.95
Seller surplus (mean)	\$16.40

Table 7: Early vs. late market comparison (single seed).

Period	Mean price	Mean liquidity
Ticks 0–4 (early)	\$84.18	54.7%
Ticks 25–29 (late)	\$86.57	61.3%

Finding 8: Market prices stabilise rapidly. The price trend slope of +\$0.057/tick is economically negligible (total drift < \$2 over 30 ticks against a mean of \$89). The absence of systematic trend is consistent with the prediction that decentralised bilateral bargaining converges to a stable price level [15], though a single-seed run without a statistical stability criterion cannot formally confirm convergence. The outcome is consistent with H-D1.

Finding 9: Law of One Price fails; price dispersion persists. Within-tick standard deviation of \$20 means that simultaneous transactions occur at substantially different prices within the same market period. This is the expected outcome under bilateral bargaining, where each pair negotiates independently and prices reflect the specific ZOPA of each pair rather than a single market-clearing price. The outcome is consistent with H-D2.

Finding 10: Buyers capture more surplus than sellers. Mean buyer surplus (\$32.95) exceeds mean seller surplus (\$16.40) by a factor of two. The mean transaction price of \$89.44 lies below the ZOPA midpoint for most pairs, consistent with a first-mover advantage: buyers open in round 0, establishing the negotiating baseline toward the lower end of the ZOPA. The outcome is consistent with H-D3, though it is observed in a single-seed run.

Finding 11: Random matching is liquidity-inefficient. Average liquidity of 52.2% means that nearly half of all matched pairs fail to trade. With random matching, many pairs are matched across non-overlapping value/cost ranges (buyer value < seller cost), making trade impossible regardless of agent capability.

Single-seed caveat. These results are from a single seed (42). The specific agent pairings, ZOPA distributions, and tick-level outcomes may vary under different seeds. The findings are suggestive but should not be treated as confirmatory.

6.3.2 Experiment E: Shock Response

Table 8 compares market outcomes under the no-shock baseline and the shock condition.

Table 8: Shock response outcomes: `no_shock` vs. `with_shock` (30 ticks each, single seed).

Metric	<code>no_shock</code>	<code>with_shock</code>	Δ
Mean price	\$90.92	\$88.77	−\$2.15
Price min	\$78.07	\$69.95	−\$8.12
Price max	\$104.76	\$100.82	−\$3.94
Within-tick price std	\$20.36	\$19.55	−\$0.81
Mean liquidity	54.2%	51.1%	−3.1 pp
Liquidity min	26.7%	13.3%	−13.4 pp
Liquidity max	86.7%	86.7%	0

Finding 12: Shocks reduce prices and liquidity. The shock condition produces a lower mean price (\$88.77 vs. \$90.92) and lower average liquidity (51.1% vs. 54.2%). Random demand and supply multipliers in $[0.7, 1.3]$ with 30% probability compress the effective ZOPA for affected pairs, reducing both the number of successful trades and the price level at which they occur. The outcome is consistent with H-E2.

Finding 13: Shocks produce asymmetric liquidity tail risk. The worst-case tick liquidity drops sharply with shocks (13.3% vs. 26.7%), while the best-case tick remains unchanged (86.7% in both conditions). Negative shocks that simultaneously suppress buyer values and inflate seller costs can eliminate nearly all viable ZOPA pairs in a single tick. Positive shocks widen the ZOPA but cannot push liquidity above the random-matching ceiling. the outcome is consistent with H-E3.

Finding 14: Price dispersion is unchanged by shocks. Within-tick price standard deviation is nearly identical across conditions (\$20.36 vs. \$19.55). The bilateral bargaining protocol—in which each pair negotiates independently—is the primary driver of price dispersion, not the width of the ZOPA. The shock mechanism shifts the ZOPA but does not change the negotiation process. H-E1 is partially rejected: while shocks affect price levels and liquidity, they do not increase within-tick price dispersion.

Single-seed caveat. As with Experiment D, these results are from a single seed. The specific shock realisations (which ticks are shocked, by how much) are fixed; different seeds would produce different shock sequences and potentially different aggregate effects.

6.3.3 Experiment I: Structural Supply-Demand Shifts

Whereas Experiment E applies stochastic shocks of random direction and magnitude, Experiment I tests deterministic, directional structural shifts: a \$20

increase in buyer values (demand shock) and a \$20 increase in seller costs (supply shock). Three conditions (**baseline**, **demand_shock**, **supply_shock**) were evaluated across 3 seeds, 10 ticks, and 10 pairs per tick, yielding 900 sessions total (300 per condition). The experimental design is described in Section 5.4.3.

Table 9: Supply-demand experiment outcomes by condition ($n = 300$ sessions per condition, pooled across 3 seeds).

Condition	Mean price	Liquidity	Total welfare	Buyer surplus	Seller surplus	Price std
baseline	\$91.81	56.0%	\$47.81	\$30.03	\$17.77	\$18.08
demand_shock	\$93.65	65.0%	\$63.83	\$45.94	\$17.90	\$20.07
supply_shock	\$108.27	26.0%	\$40.77	\$20.79	\$19.98	\$14.69

Table 10: Feasibility decomposition: fraction of pairs with positive ZOPA (mechanical), deal success among feasible pairs (behavioural), and overall liquidity.

Condition	Feasible pairs	Behavioural success	Overall liquidity
baseline	80.0%	70.0%	56.0%
demand_shock	93.7%	69.4%	65.0%
supply_shock	63.3%	41.1%	26.0%

Finding 19: Demand shock raises prices modestly. Mean price rises by \$1.84 (+2.0%), from \$91.81 under baseline to \$93.65 under the demand shock. Liquidity also increases, from 56.0% to 65.0% (+9.0 pp). Both directions are consistent with H-I1; however, the price magnitude is substantially smaller than the +17% increase in mean buyer value (\$115 to \$135), indicating attenuated demand-side price transmission.

Finding 20: Supply shock raises prices by 17.9%, closely tracking the cost shift. Mean price rises by \$16.46, from \$91.81 to \$108.27. The +\$20 cost shift therefore produces a +\$16.46 price increase, indicating near-complete but not perfect pass-through of the cost shock into transaction prices. Part of this pass-through may be gate-mediated: higher seller costs raise the feasibility floor at which the gate fires, producing mechanically higher minimum settlement prices. The relative contributions of agent-level price discovery and gate enforcement to the observed pass-through cannot be separated without gate ablation. Liquidity falls from 56.0% to 26.0% (−30.0 pp). Both directions are consistent with H-I2.

Finding 21: Supply shock collapses liquidity through both mechanical and behavioural channels. Table 10 decomposes the 30 pp liquidity decline. The fraction of mechanically feasible pairs (positive ZOPA) falls from 80.0% to

63.3%, and the deal success rate among feasible pairs falls from 70.0% to 41.1%. Both components contribute: fewer pairs can trade at all, and those that can trade succeed less often when the ZOPA is narrow. This is consistent with H-I3, though the “behavioural” channel (deal success among feasible pairs) also reflects gate involvement—narrower ZOPAs reduce the probability that the gate fires—so the agent and gate contributions within this channel are entangled.

Finding 22: Demand shock improves welfare; supply shock reduces it. Mean total welfare per session rises by 33.5% under the demand shock (\$47.81 to \$63.83), with gains accruing almost entirely to buyers: mean buyer surplus rises by 53.0% (\$30.03 to \$45.94), while mean seller surplus increases by only 0.7% (\$17.77 to \$17.90). The supply shock reduces mean total welfare per session by 14.7% (\$47.81 to \$40.77).

Finding 23: Supply shock redistributes surplus toward sellers. Mean buyer surplus falls from \$30.03 to \$20.79 (−30.8%), while mean seller surplus rises from \$17.77 to \$19.98 (+12.4%). The buyer/seller surplus ratio shifts from 1.69:1 at baseline to 1.04:1 under supply shock. Higher seller costs raise the cost floor, which directly constrains the minimum acceptable price and partially offsets the buyer first-mover advantage documented in Experiment D (Finding 10).

Demand-supply asymmetry. Symmetric \$20 shifts on opposite sides of the market produce markedly asymmetric price responses: +2.0% under the demand shock versus +17.9% under the supply shock. The mechanism driving this asymmetry is not resolved by the current experimental design and is discussed in Section 7.

Caveats. Three caveats apply. First, the mechanical and behavioural components in Table 10 are not independent: narrower ZOPAs produce both fewer feasible pairs and harder negotiation problems, so the two channels are entangled. Second, three seeds with 300 sessions per condition support directional replication claims but do not meet the threshold for formal hypothesis testing. Third, the cause of demand-side price attenuation is unresolved; possible explanations include bounded rationality in offer adjustment, anchoring to baseline price distributions, or model-specific response patterns that are not separable without ablation.

6.3.4 Experiment H: Institutional Mechanism Comparison

Table 11 compares market outcomes under two matching mechanisms: random pairing and surplus-maximising (greedy maximum-ZOPA) pairing.

The surplus-maximising matcher produces fewer total sessions (220 vs. 300) because it only pairs agents with positive ZOPA overlap, leaving unprofitable pairs unmatched. Despite fewer sessions, it achieves nearly as many deals (158 vs. 167) and a substantially higher deal rate (71.8% vs. 55.7%).

Table 11: Mechanism comparison outcomes (10 ticks $\times$ 10 pairs $\times$ 3 seeds per condition).

Metric	random	surplus_max
Total sessions	300	220
Deals made	167	158
Deal rate	55.7%	71.8%
Mean price	\$91.47	\$87.24
Price std	\$18.60	\$16.45
Allocative efficiency	83.9%	89.7%
Surplus Gini	0.226	0.216
Buyer surplus (mean)	\$30.54	\$42.25
Seller surplus (mean)	\$17.39	\$16.17

Finding 15: Surplus-maximising matching improves deal rates by 16 percentage points. The `surplus_max` condition’s deal rate of 71.8% exceeds the `random` condition’s 55.7%. By filtering out pairs with non-overlapping or narrow ZOPAs before negotiation begins, the mechanism ensures that matched pairs have sufficient surplus for the LLM agents to reach agreement. H-H1 is supported.

Finding 16: Mechanism design improves allocative efficiency. Allocative efficiency rises from 83.9% under random matching to 89.7% under surplus-maximising matching, a 5.8 percentage point improvement. The mechanism captures more of the theoretically available welfare by directing trades toward high-surplus pairs. H-H2 is supported.

Finding 17: Buyer surplus increases disproportionately under surplus-maximising matching. Mean buyer surplus rises from \$30.54 to \$42.25 (+38%), while seller surplus is essentially unchanged (\$17.39 vs. \$16.17). Under surplus-maximising pairing, the additional surplus available from wider ZOPAs accrues predominantly to buyers, consistent with the first-mover advantage observed in Experiment D.

Finding 18: Surplus-maximising matching reduces price dispersion. Price standard deviation falls from \$18.60 to \$16.45. By excluding low-overlap pairs that produce extreme prices, the mechanism narrows the price distribution, though within-session dispersion remains substantial.

6.4 Cross-Experiment Synthesis

Tables 12 and 13 synthesise the empirical patterns observed across all seven experiments (A–E, H, I).

Table 12: Where platform outcomes are consistent with economic theory. The “Primary driver” column indicates whether the outcome depends primarily on the agent’s decisions (Agent), the negotiation protocol and constraint structure (Protocol), the feasibility gate (Gate), or the matching mechanism design (Mechanism). Findings marked [†] are from single-seed exploratory runs.

Economic principle	Evidence	Exp	Primary driver
ZOPA determines deal success	2.7% vs. 100% deal rate as ZOPA narrows	B	Protocol/Gate
Market price convergence [†]	+\$0.057/tick slope; stable \$89 mean	D	Protocol/Gate
Bilateral bargaining dispersion [†]	Within-tick $\sigma = \$20$; Law of One Price fails	D	Protocol
First-mover buyer advantage [†]	Buyer surplus \$32.95 > seller surplus \$16.40	D	Protocol
Shock $\rightarrow$ price/liquidity reduction [†]	−\$2.15 price, −3.1 pp liquidity	E	Protocol/Gate
Asymmetric shock tail risk [†]	Worst-case liquidity drops 13.4 pp; best-case unchanged	E	Protocol/Gate
Mechanism $\rightarrow$ efficiency gain	+5.8 pp allocative efficiency under surplus-max	H	Mechanism
Mechanism $\rightarrow$ deal rate gain	+16.1 pp deal rate under surplus-max matching	H	Mechanism
Supply shock $\rightarrow$ price increase	+\$16.46 (+17.9%) under +\$20 seller cost shift	I	Protocol/Gate
Demand shock $\rightarrow$ welfare gain	+33.5% total welfare; gains accrue to buyers	I	Protocol/Agent
Supply shock redistributes surplus	Surplus ratio shifts from 1.69:1 to 1.04:1	I	Protocol/Gate

Table 13: Where LLM agents depart from classical predictions.

Predicted behaviour	Observed	Interpretation
Anchoring bias	$r = -0.148$	Prompt provides explicit surplus
Deadline concession	Closes at round 3 always	Gate + prefer-counter steering
Shocks increase dispersion	σ unchanged	Protocol drives dispersion, not agent
Demand shock $\rightarrow$ proportional price increase	+2.0% observed vs. +17% input shift	Demand-side response attenuated

Feasibility gate as cross-cutting confound. The gate drives approximately 40% of all acceptances. In fixed-scenario experiments (A–C), it fires at round 2 in every session, limiting concession data to two observations and rendering settlement timing invariant to deadline manipulation. In market experiments (D, E, I), gate involvement varies by session depending on the drawn ZOPA. The gate’s contribution to deal rates and settlement timing cannot be separated from agent capability without ablation studies that disable the gate entirely.

First-mover advantage as protocol effect. Buyer surplus exceeds seller surplus in both the fixed-scenario concession experiment (A) and the market experiment (D). This pattern is consistent with the buyer-first turn order in the alternating-offer protocol: the buyer’s opening offer anchors the negotiation toward the lower end of the ZOPA. This is a protocol effect, not an agent-type effect, and would persist regardless of the LLM used. Experiment I (Finding 23) shows that higher seller costs partially offset this advantage: the buyer/seller surplus ratio narrows from 1.69:1 at baseline to 1.04:1 under supply shock, indicating that the first-mover effect is context-sensitive rather than unconditional.

Prompt steering as uncontrolled variable. The prefer-counter instruction and the accept-condition block are active in all LLM conditions across all experiments. Their combined effect is to reduce rejection, accelerate settlement, and possibly suppress anchoring. Because neither has been ablated, they function as uncontrolled confounds rather than experimental variables. Separating their effects from the model’s intrinsic bargaining behaviour is an important direction for future work.

Demand-supply asymmetry as cross-experiment pattern. Across Experiments E and I, supply-side cost shocks produce stronger and more consistent price and liquidity responses than demand-side value changes. In Experiment E, stochastic shocks that compress the ZOPA reduce both prices and liquidity. In Experiment I, a directed supply shock produces a near-proportional price increase (+17.9% from a +\$20 cost shift) while a symmetric demand shock produces a muted price response (+2.0% from a +\$20 value increase). The mechanism driving this asymmetry has not been isolated; possible explanations include the structural primacy of cost floors as hard constraints, differential prompt sensitivity to cost versus value information, or nonlinear ZOPA effects. Confirming any interpretation requires targeted ablation experiments.

7 Discussion

This section interprets the experimental results, identifies the principal methodological tensions, situates the findings relative to prior work, and identifies the limitations of the current study.

7.1 What the Simulator Successfully Demonstrates

The experimental programme reveals a consistent set of outcomes that the platform can credibly demonstrate: market-level regularities that are structurally determined by the negotiation protocol, private constraints, and the feasibility gate.

On the structural side, agents reliably produce ZOPA-determined deal success (Experiment B), stable market prices with persistent within-tick dispersion (Experiment D), first-mover surplus advantage (Experiments A and D), directionally correct responses to supply and demand shocks (Experiments E and I), and mechanism-driven efficiency gains (Experiment H). These outcomes depend primarily on the existence and width of the zone of possible agreement, the alternating-offer protocol, and the matching mechanism—not on whether the agent reasons strategically. A trivially rational agent that accepts any positive-surplus offer would produce qualitatively similar results in wide-ZOPA scenarios, though the specific price paths would differ. The gate, which implements exactly this logic, drives approximately 40% of all acceptances.

Experiment H demonstrates that the matching mechanism—an institutional design choice external to the agents—has a substantial effect on market out-

comes. Surplus-maximising matching improves deal rates by 16 percentage points and allocative efficiency by 5.8 percentage points relative to random matching, while also reducing price dispersion. These gains arise entirely from pair selection: the mechanism filters out low-ZOPA pairs before negotiation begins, ensuring that LLM agents face problems they can solve. The surplus-maximising matcher is an oracle mechanism—it requires knowledge of both parties’ private constraints—but it establishes an upper bound on the efficiency gains achievable through mechanism design in LLM-agent markets.

The asymmetric surplus distribution under surplus-maximising matching—buyer surplus rises 38% while seller surplus is essentially unchanged—suggests that mechanism design interacts with the first-mover protocol. Wider ZOPAs created by better matching amplify the buyer’s first-mover advantage rather than distributing surplus equally. This has implications for mechanism designers seeking both efficiency and equity in automated negotiation markets.

Experiment I provides the clearest supply-side confirmation: a +\$20 cost shift produces near-complete price pass-through (+17.9%), replicated across three seeds. The welfare and surplus redistribution results are directionally consistent with supply-demand theory. Under supply shock, the buyer/seller surplus ratio narrows from 1.69:1 (\$30.03 vs. \$17.77 at baseline) to 1.04:1 (\$20.79 vs. \$19.98), consistent with sellers passing through higher costs under a compressed ZOPA.

Taken together, these positive findings indicate that the platform can serve as a controlled testbed for evaluating how protocol and institutional structure shape market outcomes, even when individual agent reasoning is limited.

7.2 Why Some Experiments Fail or Become Ambiguous

Several experiments produce null results or results too confounded to interpret cleanly. These failures share a common source: design decisions that were engineering necessities introduced confounds that prevent clean causal attribution.

The gate as confound. The feasibility gate fires at round 2 in every fixed-scenario session, limiting concession data to two observations per session and making settlement timing invariant to deadline manipulation. This is the primary reason Experiment C produces a null result: deals settle before deadline proximity becomes relevant. Extended concession trajectories are unobservable, and 100% deal rates in wide-ZOPA scenarios may reflect gate capability rather than agent capability. Without gate ablation, agent capability cannot be separated from gate enforcement.

Prompt steering confound. The prompt instruction to prefer COUNTER over REJECT removes the primary mechanism by which agents could delay settlement. In combination with the gate, this instruction makes deadline sensitivity structurally impossible to observe in the current design. The instruction was introduced for a practical reason: without it, small language models frequently

reject profitable offers, producing meaningless 0% deal rates. This illustrates a recurring tension between engineering reliability and experimental cleanliness.

The anchoring null. The accept-condition block pre-computes the utility of the opponent’s standing offer and presents it directly in the prompt. In human anchoring experiments [5], the anchoring effect depends on negotiators forming impressions under uncertainty about the opponent’s reservation price. Our agents receive explicit surplus information, potentially overriding any tendency to anchor on the first offer. This is a plausible explanation for the anchoring null result ($r = -0.148$, Experiment B), but it has not been tested: an ablation that removes the accept-condition block would be needed to confirm or reject this hypothesis. The independent variable in Experiment B also simultaneously manipulates ZOPA width and first-offer level, meaning anchoring cannot be fully isolated from ZOPA feasibility effects.

Demand-side attenuation. Experiment I identifies a striking asymmetry: symmetric \$20 shifts to buyer values and seller costs produce price responses of +2.0% and +17.9% respectively. One structural explanation is that seller cost functions as a hard lower bound on acceptable prices, embedded in both the gate and the agent’s prompt-level constraints, producing near-mechanical cost pass-through. Buyer value, by contrast, is a soft upper bound that buyers do not fully exploit: in Experiment A, reactive buyers open at \$65.46 against a \$120 value (Table 3), and the first-mover protocol means this conservative opening anchors the negotiation downward. A value increase raises the ceiling that buyers already do not approach, producing an attenuated price response. Alternative explanations include ZOPA-mediated mechanical effects and model-specific prompt response patterns. The three-seed directional replication strengthens confidence in the observed patterns but does not resolve the mechanism question.

The supply shock also reveals that LLM agent effectiveness degrades as the bargaining problem becomes harder. Table 10 shows that the deal success rate among mechanically feasible pairs falls from 70.0% to 41.1% under supply shock, indicating that narrow ZOPAs impair LLM convergence even when agreement is structurally possible. This parallels the finding in Experiment B that the model achieves only a 2.7% deal rate when the ZOPA narrows to \$20 (Table 4). Both patterns point to the same qualitative conclusion: LLM-agent market liquidity is doubly sensitive to cost shocks, through both the mechanical reduction in feasible pairs and the behavioural degradation among those that remain.

Prompt architecture as cognitive architecture. In human negotiation research, the cognitive architecture of the bargainer is relatively fixed; experimental variation comes from the task, the opponent, and the information structure. In LLM negotiation, the prompt *is* the cognitive architecture, and any change—adding a reasoning scaffold, including or excluding surplus information, steering toward or away from rejection—can alter outcomes. LLM negotiation results are therefore inherently prompt-conditional. The deliberative prompt’s

two-step reasoning scaffold (“Is the offer acceptable? If not, what price?”) leads the model to hold a firmer position on its first counter (\$7.21 difference, Experiment A), consistent with Abdelnabi et al. [1], but whether this reflects genuine multi-step reasoning or simply a constrained output distribution cannot be determined from the current experiments.

7.3 Micro–Macro Tension

A structural tension runs through the entire experimental programme: designs that preserve raw agent behaviour may weaken aggregate market regularity, while stronger enforcement can stabilise outcomes but partially replaces agent decision-making.

In the micro-level experiments (A–C), the feasibility gate is enabled and fires at round 2 in every session. This produces 100% deal rates in wide-ZOPA conditions and allows clean comparison across prompt conditions. The cost is that the multi-round concession dynamics the experiments were designed to study are never observed. Between the model’s generation and the enacted action, three successive mediation layers operate: the rethink loop corrects format and self-constraint violations inside the agent, the ActionJudge enforces protocol and global-bound constraints at the session level, and the gate (when enabled) overrides the agent’s decision entirely. The aggregate outcome metric—deal rate—is stable and comparable, but it is the gate’s outcome, not the agent’s.

In the market-level experiments (D, E, H, I), the gate continues to enforce acceptance of clearly profitable offers. This is sufficient to generate observable market structure—price convergence, surplus asymmetry, shock response—but it means that market liquidity is partly gate-enforced rather than emergent. The counterfactual without the gate can be inferred from Experiment B’s narrow-ZOPA condition, where the gate cannot fire and the model fails 97.3% of the time. A gate-disabled market experiment would likely produce near-zero deal rates with a 3B-parameter model, making market-level analysis impossible.

This reveals a fundamental constraint on LLM simulation research with small models: the researcher must choose between experimental cleanliness (gate off, agent behaviour is unmediated, but the market may not function) and practical viability (gate on, market functions, but agent behaviour is partially replaced by enforcement logic). There is no design that fully satisfies both requirements at once for a 3B-parameter model in alternating-offer bargaining.

The tension is not unique to this platform. Any structured economic simulation using LLM agents must decide how much enforcement to impose to make the experiment function. The contribution of the current work is to make this trade-off explicit, quantify the gate’s involvement rate (approximately 40% of acceptances globally), and log every gate-fired settlement separately—allowing readers to assess the extent of enforcement in any given experiment. This transparency is a prerequisite for any future comparison across model sizes, where larger models may require less enforcement and the gate involvement rate may fall.

7.4 Comparison to Prior Work

We compare our findings to the prior work reviewed in Section 2.

Generative agent simulations. Park et al. [11] demonstrated that LLM agents exhibit emergent social behaviours in open-ended environments. Our market experiments (D and E) extend this paradigm to economic settings: LLM agents produce emergent market structure—stable price equilibrium, persistent price dispersion, and surplus asymmetry—without being explicitly programmed to produce these aggregate outcomes. The key difference is that our environment is constrained (hard budget and cost limits, structured protocol) rather than open-ended, enabling direct comparison against theoretical predictions.

LLM negotiation. Abdelnabi et al. [1] find that deliberation prompting improves agreement quality in multi-party, multi-issue negotiation games. Our Experiment A provides a complementary finding in a simpler setting: deliberative prompting produces more conservative counter-offers (\$7.21 difference), consistent with improved strategic positioning, though we observe it through a different metric (offer price rather than Pareto efficiency). Our setting is single-issue bilateral bargaining, which isolates price dynamics from the issue-trading effects present in their multi-issue design.

Mao et al. [9] show that personality-assigned prompts shift negotiation outcomes. Our prefer-counter instruction functions similarly as a prompt-level modifier. The platform additionally implements communication-strategy modifiers (assertive, collaborative, strategic) that are analogous to Mao et al.’s personality prompts, but these were not activated in the reported experiments; all runs used the default neutral strategy. The methodological difference is that we disclose prompt steering as a confound rather than studying it as a controlled variable.

Lewis et al. [8] found that end-to-end trained negotiation agents develop non-human-like strategies, including degenerate repetitive behaviour. Our LLM agents show a different form of departure from human norms: rather than degenerating, they fail to reproduce specific human heuristics (anchoring, deadline pressure) while maintaining otherwise reasonable bargaining behaviour. Both findings suggest that artificial agents do not spontaneously replicate the full spectrum of human negotiation behaviour, though the underlying mechanisms differ—reinforcement learning via self-play versus prompt-conditioned inference.

Classical theory. Our positive results confirm several qualitative predictions from bilateral bargaining theory. Market prices stabilise rapidly and exhibit persistent within-tick dispersion (consistent with Rubinstein and Wolinsky [15]), ZOPA width determines deal feasibility (consistent with Raiffa [12]), and the buyer-first protocol produces buyer-favourable surplus splits (consistent with first-mover advantage in alternating-offer models [14]). Experiment I strengthens the supply-side confirmation with a three-seed directional replication: a +\$20 cost shift produces near-complete price pass-through (+17.9%). The

demand-side confirmation is weaker: a symmetric +\$20 value shift produces only a +2.0% price increase, directionally correct but substantially attenuated relative to the input perturbation.

Our null results depart from two well-established human findings. The absence of anchoring bias contradicts Galinsky and Mussweiler [5], though the departure is plausibly explained by the prompt’s provision of explicit surplus information—a condition that does not hold in human experiments. The absence of deadline sensitivity contradicts Roth et al. [13], though this result is heavily confounded by the feasibility gate and prompt steering, and should not be interpreted as evidence that LLM agents lack the capacity for deadline awareness under different experimental conditions.

7.5 Limitations

We identify eleven limitations of the current study, ordered by their impact on the interpretation of results.

1. **Single model.** All results are specific to `llama3.2:3b` at temperature 0.2. Larger models, different architectures, or fine-tuned models may exhibit anchoring bias, deadline sensitivity, or other behaviours absent in our experiments. No cross-model comparison is provided.
2. **Gate dominance.** In wide-ZOPA scenarios, the feasibility gate drives all acceptances. Deal rates and settlement timing in Experiments A–C reflect gate logic rather than agent decision-making. Without gate ablation, agent capability cannot be separated from gate enforcement. Furthermore, even in gate-disabled conditions, the rethink loop remains active and partially pre-filters the action space by correcting self-constraint violations before they reach the ActionJudge; gate-off configurations therefore do not expose fully unmediated LLM output.
3. **Two-round data.** The gate fires at round 2 in all fixed-scenario sessions, yielding only two data points per session for concession analysis. Multi-round concession dynamics—the core phenomenon the concession experiment was designed to study—are not observed.
4. **No prompt ablation.** The accept-condition block, prefer-counter steering, and deadline-salience language are all active but none are individually ablated. Their separate effects on anchoring, deal rates, and deadline sensitivity are unknown.
5. **Single seed for market experiments.** Experiments D and E use seed 42 only. The specific agent pairings, ZOPA distributions, and shock realisations are fixed. Results are suggestive but not confirmatory.
6. **Confounded anchoring design.** The independent variable in Experiment B simultaneously manipulates ZOPA width and first-offer level. The Pearson r metric partially addresses this, but anchoring cannot be fully isolated from ZOPA feasibility effects.

7. **No statistical tests.** All analyses are descriptive (means, standard deviations, Pearson r). No hypothesis tests, confidence intervals, or effect sizes are reported. The three-seed design in Experiments A–C provides limited replication; Experiments D–E have no replication.
8. **Single-price negotiation.** All sessions negotiate a single price. Real markets involve quality, delivery terms, and bundling. Multi-issue negotiation may produce qualitatively different dynamics, including the logrolling and issue-trading effects studied by Lewis et al. [8].
9. **No cross-run learning.** Agents do not accumulate experience across sessions within the completed experiments. The memory and reputation agents are implemented but untested. Whether LLM agents adapt strategies over repeated interactions remains an open question.
10. **LLM non-determinism.** Even at temperature 0.2, model outputs are not fully deterministic. The seeded RNG controls market-level randomness (population generation, matching, shocks) but not LLM inference randomness. This introduces uncontrolled variance into all LLM-agent conditions.
11. **Demand-supply asymmetry mechanism unresolved.** The demand-supply price asymmetry in Experiment I (symmetric \$20 shifts producing +2.0% versus +17.9% price responses) is documented but not mechanistically explained. Three candidate explanations—hard cost-floor constraints, conservative buyer first-mover positioning, and nonlinear ZOPA sensitivity—are consistent with the data but have not been isolated through targeted ablation.

8 Conclusion and Future Work

8.1 Summary

This thesis presented a controlled experimental platform for evaluating LLM bargaining behaviour in alternating-offer bilateral negotiation under private economic constraints. The platform enforces hard budget and cost limits, provides a configurable deterministic settlement gate, logs every agent action at sufficient granularity to separate LLM-generated decisions from mechanically enforced settlements, and connects micro-level bargaining behaviour to macro-level market outcomes within a single system. Using this platform, we conducted seven experiments totalling 4,120 negotiation sessions with a single 3B-parameter language model (llama3.2:3b).

The experiments produced two categories of findings. First, platform-level outcomes were consistent with several predictions of classical bargaining and market theory: deal success scaled with the width of the zone of possible agreement, market prices showed no systematic trend in single-seed exploratory

runs, the buyer-first protocol produced systematic surplus asymmetry, stochastic shocks reduced prices and liquidity in predicted directions, structural supply-demand shifts produced directionally correct price responses with near-complete cost pass-through on the supply side, and surplus-maximising matching improved deal rates and allocative efficiency relative to random matching. These outcomes arise primarily from the negotiation protocol, private constraints, and the feasibility gate rather than from agent sophistication—a 3B-parameter model, assisted by a gate that drives approximately 40% of acceptances globally (and 100% in the micro-level experiments), is sufficient to produce qualitatively correct market structure when the gate enforces acceptance of profitable offers.

Second, agents did not exhibit two well-documented human bargaining heuristics: anchoring bias and deadline-driven concession acceleration. These null results are heavily confounded by the experimental design. The anchoring null may reflect the prompt’s provision of explicit surplus information, but the gate also settles all wide-ZOPA sessions at round 2, leaving no multi-round trajectory on which anchoring could manifest. The deadline null is confounded by the gate (which settles deals before deadlines approach) and by prompt instructions that discourage rejection. Neither confound has been ablated; the null results characterise the combined system (LLM + gate + prompt) under these specific conditions, and the current design cannot distinguish LLM cognitive limitations from gate-induced truncation.

8.2 Implications

The findings carry implications for three audiences.

For automated negotiation deployment. LLM agents settle bilateral deals reliably when the ZOPA is wide, but fail at a 97.3% rate when the ZOPA narrows to \$20 (Experiment B). Practical systems that deploy LLM negotiators should estimate ZOPA width before entering negotiation and provide deterministic fallbacks—analogue to the feasibility gate—for scenarios where the model is likely to fail. The absence of anchoring bias may be advantageous in adversarial settings (the agent is not manipulated by aggressive first offers), though it also means the agent cannot exploit anchoring strategically.

For experimental methodology. The feasibility gate illustrates a general tension in LLM agent research: engineering interventions that make experiments viable can simultaneously mask the behaviours those experiments are designed to study. Researchers using LLM agents in structured settings should treat such interventions as controllable variables, report their involvement rates, and conduct ablation studies to separate agent capability from system-level enforcement. Prompt architecture is a first-order experimental variable; LLM negotiation results should be understood as prompt-conditional, and claims about LLM negotiation behaviour in general should be hedged accordingly.

For economic theory. The positive results suggest that certain market-level predictions of bilateral bargaining theory—price convergence, price dispersion, first-mover advantage, directional shock response—may hold even when agents lack sophisticated strategic reasoning, provided enforcement mechanisms ensure acceptance of clearly profitable offers. These properties emerge from the structure of bilateral bargaining itself (independent pairwise negotiation, private constraints, alternating offers) and from the gate, rather than from agent sophistication. This is a modest finding: it demonstrates that the structural predictions of bilateral bargaining theory hold under a relatively weak rationality assumption—that agents accept offers yielding positive surplus when explicitly informed of that surplus—not that LLM agents are economically rational or can substitute for human participants in experimental economics.

8.3 Future Work

Several directions would strengthen and extend the current findings.

Gate and prompt ablation. The most immediate priority is to repeat Experiments A–C with the gate disabled and to separately ablate the accept-condition block, prefer-counter steering, and deadline-salience language. This would reveal which null results are gate artefacts and which reflect intrinsic LLM behaviour. The platform already supports gate toggling via a single configuration flag (`gate_enabled`), making this ablation straightforward to execute.

Multi-model comparison. Repeating key experiments with models of different sizes (7B, 13B, 70B) and architectures would test whether anchoring bias and deadline sensitivity emerge at larger scales. This is the single most important external-validity question: whether the null results are specific to a 3B-parameter model or whether they generalise across model families.

Multi-seed market experiments. Experiments D and E should be replicated across multiple seeds to assess whether the market-level findings (price convergence, surplus asymmetry, shock response) are robust or specific to the seed-42 random matching.

Supply-demand mechanism isolation. Experiment I establishes the asymmetric price response to symmetric cost and value shifts but cannot isolate the responsible mechanism. Three targeted ablations would address this: (i) running the demand-shock condition with the gate disabled to test whether demand attenuation is a gate artefact; (ii) varying the size of the demand shift to test whether the price response is linear or threshold-gated; (iii) running the supply-shock condition with different first-mover protocols (seller-first or simultaneous) to test whether cost-floor pass-through depends on the buyer-first turn order.

Reputation and communication experiments. The platform supports two additional experiment families—Experiment F (reputation and repeated interaction using the round-robin matcher and reputation agent) and Experiment G (communication strategy ablation using prompt-level communication modifiers)—for which infrastructure is implemented but empirical evaluation has not been conducted. Both experiments require the gate to be disabled so that strategic adaptation can manifest over multiple rounds.

Statistical analysis. Formal hypothesis tests, confidence intervals, and effect-size measures would strengthen the evidential basis of the current findings. The three-seed design in Experiments A–C supports seed-level analysis of variance; the single-seed market experiments require additional replication before statistical testing is meaningful.

Human–LLM comparison. Running the same experimental conditions with human participants would enable direct comparison of bargaining behaviour. This would clarify whether the absence of anchoring and deadline effects is specific to LLMs or whether it extends to any agent operating with explicit constraint information.

Stronger baselines. The rule-based agent provides a deterministic baseline but follows a simple linear concession strategy. Adding stronger algorithmic baselines—such as Nash-equilibrium-computing agents or game-theoretic optimal strategies—would help calibrate the gap between LLM behaviour and theoretically optimal play.

The platform is a testbed, not a finished product. Its value lies in enabling systematic, reproducible comparison of LLM bargaining behaviour against well-characterised theoretical predictions—including honest reporting of where that behaviour aligns with theory and where it does not.

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